

Jian Sun

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Educational Experience

M.S. in Engineering Mechanics *September 2020 - June 2023*
University of Chinese Academy of Sciences, Institute of Mechanics

Relevant Coursework: Finite Element Methods, Nonlinear Finite Element Methods, Fracture Mechanics, Numerical Analysis, Master's Degree English

B.S. in Engineering Mechanics *September 2013 - June 2017*
Lanzhou University of Technology

Relevant Coursework: Advanced Mathematics, Linear Algebra, Probability Theory and Mathematical Statistics, Theoretical Mechanics, Mechanics of Materials, Elastic Mechanics

Project Experience

Modeling and Analysis of High-Cycle Fatigue Performance
Key Research Project *November 2021 - February 2022*

- Used ABAQUS software to model and analyze high-cycle fatigue samples, calculating stress concentration factors under different loading conditions.
- Applied solid mechanics knowledge to compute the control volume and control surface area sizes within the 90% stress region of fatigue samples.
- Developed MATLAB programs to predict P-S-N curves for samples of different sizes using Weibull distribution based on experimental data.

Sub-Project on Gear Steel Contact Fatigue
Pinion steel Project *March 2022 - June 2022*

- Modeled and simulated the stress distribution at contact points of contact fatigue components using ABAQUS, comparing results with theoretical calculations.
- Extended the analysis from 2D to 3D modeling to determine stress distribution in three-dimensional contact problems.
- Communicated with domain experts to validate simulation results, providing theoretical support and practical guidance for subsequent gear steel contact fatigue experiments.

Publications and Conferences

- Sun, Jian & Peng, Wenjie & Sun, Chengqi. (2022). Mechanism of artificial surface defect induced cracking for very high cycle fatigue of Ti alloys. *Engineering Fracture Mechanics*. 272, 108721.
- Sun, Jian & Wu, Lei & Sun, Chengqi. (2021). Effects of Notches and Defects on Dwell Fatigue Mechanism and Fatigue Life of Ti-6Al-4V ELI Alloy Used in Deep-Sea Submersibles. *Journal of Marine Science and Engineering*. 9, 845.
- Li, Gen, Liu, Jiajun, Sun, Jian & Sun, Chengqi. (2023). Effects of Natural Aging and Discontinuous Cyclic Loading on High Cycle Fatigue Behavior of Steels. *Metals*. 13(3), 511.
- Sun, Chengqi, Sun, Jian, Chi, Weiqian, Wang, Jiaxuan & Wang, Wenjing. (2023). A method of quasi in-situ EBSD observation for microstructure and damage evolution in fatigue and dwell fatigue of Ti alloys. *International Journal of Fatigue*. 176, 107897.

Awards

- Excellent Award in the 2022 English Translation Competition at the University of Chinese Academy of Sciences, with a prize of 500 RMB.
- First and Third Prize in the University Mechanics Competition at Lanzhou University of Technology.
- Third Prize Scholarship for the 2014-2015 academic year at Lanzhou University of Technology.
- Fourth place in the Men's Group A table tennis team event at the Third Gansu University Games.
- Fifth place in the Men's Team event at the 16th "College Cup" Table Tennis Tournament in Gansu Province.
- Served as President of the "Physics and Mechanics Enthusiast Association", leading the association to win the "Top Ten Campus Associations" title.

Contact Information

University of Chinese Academy of Sciences, Institute of Mechanics

Research Direction: Dwell Fatigue of Titanium Alloys

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